PHYSICS
Stage 6

Module 1: Kinematics
Motion in a Straight Line
Motion on a Plane

Teacher: Dean Dimarti
Exam Equivalent Time: 63 minutes *(based on allocation of 1.5 minutes per mark)*

Blackwattle Bay

Questions

1. PHYSICS, M1 EQ-Bank 10356

A truck travelling in a straight line with a speed of 60 ms^{-1} slows down and comes to rest over a period of 20 seconds.

a. Calculate the change in velocity of the truck. *(2 marks)*

b. Determine the average acceleration of the truck across the 20 seconds. *(2 marks)*

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2. PHYSICS, M1 EQ-Bank 10089

A ball is thrown vertically upwards from ground level, it gains 50 metres vertically and then falls back to the ground.

a. Determine the initial velocity, in metres per second, that the ball was thrown. Give your answer correct to one decimal place. *(2 marks)*

b. Calculate the total time of flight of the ball, in seconds, giving your answer correct to two decimal places. *(2 marks)*

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3. PHYSICS, M1 EQ-Bank 11246

A boat is trying to cross a river that is 200 meters wide. The boat's speed in still water is 5 m/s. The river flows eastward with a current of 3 m/s.

a. If the boat heads straight north (perpendicular to the current), using a vector diagram. determine the magnitude and direction of its velocity relative to the ground? *(3 marks)*

b. How far downstream (east) will the boat have drifted by the time it reaches the opposite bank? *(1 mark)*

c. At what angle should the boat be aimed (relative to north) so that it reaches directly across from its starting point? *(2 marks)*

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4. PHYSICS, M1 EQ-Bank 10534

A motorbike is travelling to the east at 50 ms^{-1} when it passes a car travelling west at 60 ms^{-1}.

Calculate the velocity of the motorbike relative to the car. *(2 marks)*

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5. PHYSICS, M1 EQ-Bank 10979

A bus departs from its depot, starting from rest and accelerating uniformly at 2.0 ms^{-2} for 10 seconds until it reaches a speed of 20 ms^{-1}. It then travels at this constant speed for 50 seconds before decelerating uniformly at – 2.5 ms^{-2} until coming to a complete stop at the next bus stop.

a. In the space provided, sketch a velocity–time graph of the bus's journey, clearly marking the acceleration phase, the constant speed phase, and the deceleration phase with all key values. *(3 marks)*

b. Calculate the total distance the bus covers during this trip. *(2 marks)*

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6. PHYSICS, M1 EQ-Bank 11157

A motorboat was travelling 15 m/s relative to the water, heading due east according to its navigation system.

The boat then experiences a river current flowing south at 5 m/s relative to the shore. Using a vector diagram, determine the boat's resultant velocity relative to the shore. *(2 marks)*

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7. PHYSICS, M1 EQ-Bank 10801

A car is travelling north and approaching an intersection at 50 kmh^{-1}.

While maintaining a constant speed, the car turns left and continues east at 50 kmh^{-1}.

Using a vector diagram, calculate the change in velocity of the car. *(3 marks)*

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8. PHYSICS, M1 EQ-Bank 11602

A helicopter is cruising with an airspeed, v_1, of 220 km/h. The pilot wishes for its flight path to be due south. However, a crosswind, v_2, of 60 km/h is blowing from the west.

Using a vector diagram, in which direction should the pilot steer the helicopter to achieve a due south course? *(3 marks)*

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9. PHYSICS, M1 EQ-Bank 10623

Below is a description of the motion of a runner. The motion can be divided into three stages.

Stage 1: Runner travels 120 metres south taking 20 seconds.

Stage 2: Runner turns west and travels at 5 ms^{-1} for half a minute.

Stage 3: Runner travels directly back to their starting position.

a. Determine the distance that the runner ran during Stage 2 of their journey. *(1 mark)*

b. Find the displacement of the start point from the runner Stage 2 is completed. *(3 marks)*

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10. PHYSICS, M1 EQ-Bank 11513

Two boats begin their journey from the same dock at the same time.

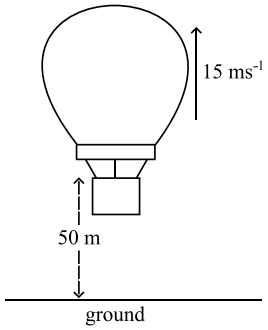
- Boat A travels at x m/s due north.
- Boat B travels at y m/s due east.

Determine whether the magnitude of the velocity of Boat A relative to Boat B would change if both boats doubled their speeds but continued in their original direction. Explain your reasoning. *(2 marks)*

11. PHYSICS, M1 EQ-Bank 10267

A hot-air balloon is travelling at a constant upwards velocity of 15 ms^{-1}.

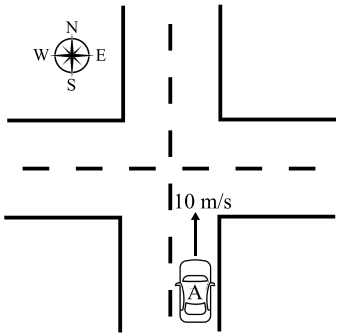
A passenger on the hot-air balloon decides to time how long it takes a pen to hit the ground when dropped from a height of 50 m.



Ignoring air resistance, determine how long it will take the pen to hit the ground. *(4 marks)*

12. PHYSICS, M1 EQ-Bank 10445

Car A approaches an intersection at 10 ms^{-1} from the South as shown. As Car B approaches the intersection, it measures the velocity of Car A to be 22.36 ms^{-1} NW.



Draw Car B on the diagram showing its direction and speed. Show all working. *(3 marks)*

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Worked Solutions

1. PHYSICS, M1 EQ-Bank 10356

a. $\Delta v = v - u$

$= 0 - 60$

$= -60 \text{ ms}^{-1}$

b. $a = \dfrac{v-u}{t}$

$= \dfrac{-60}{20}$

$= -3 \text{ ms}^{-2}$

2. PHYSICS, M1 EQ-Bank 10089

a. At 50 m $\Rightarrow$ $v = 0 \text{ ms}^{-1}$

$v^2 = u^2 + 2as$

$0^2 = u^2 + 2 \times -9.8 \times 50$

$u^2 = 980$

$u = 31.3 \text{ ms}^{-1}$

b. Time to highest point $= \dfrac{1}{2}$ time of flight

$v = u + at$

$0 = 31.3 - 9.8t$

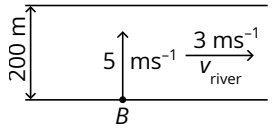
$9.8t = 31.3$

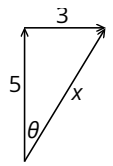
$t = 3.194 \text{ s}$

Total time of flight $= 2 \times 3.194 = 6.39 \text{ s (2 d.p.)}$

3. PHYSICS, M1 EQ-Bank 11246

a. Using the vector diagrams below



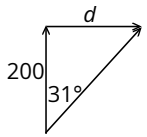


$x = \sqrt{3^2 + 5^2} = 5.83 \text{ ms}^{-1}$

$\tan\theta = \frac{3}{5} \quad \Rightarrow \quad \theta = \tan^{-1}\left(\frac{3}{5}\right) = 31^\circ$

$\therefore$ The velocity of the boat relative to the ground is 5.83 m/s N31°E.

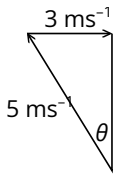
b.



$\tan 31^\circ = \frac{d}{200}$

$d = 200 \times \tan 31^\circ = 120 \text{ m}$

c.



$\sin\theta = \frac{3}{5} \quad \Rightarrow \quad \theta = \sin^{-1}\left(\frac{3}{5}\right) = 36.9^\circ$

$\therefore$ The boat should head N36.9°W.

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4. PHYSICS, M1 EQ-Bank 10534

Let east be the positive direction and west be the negative direction.

$$\begin{aligned} v_{\text{m rel c}} &= v_{\text{m}} - v_{\text{c}} \\ &= 50 - (-60) \\ &= 110 \text{ ms}^{-1} \end{aligned}$$

Therefore, the velocity of the motorbike relative to the car is 110 ms^{-1} east.

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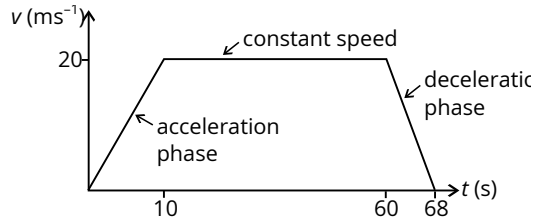
5. PHYSICS, M1 EQ-Bank 10979

a.



b. Total distance travelled by the bus is equal to the area under the velocity-time graph.

$$\begin{aligned} \text{Distance } &= (\frac{1}{2} \times 10 \times 20) + (50 \times 20) + (\frac{1}{2} \times 8 \times 20) \\ &= 1180 \text{ m} \end{aligned}$$

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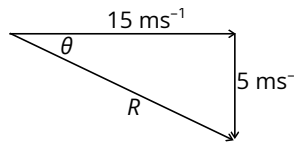
6. PHYSICS, M1 EQ-Bank 11157



Using vector addition as shown in the diagram above:

$R = \sqrt{15^2 + 5^2} = 15.8 \text{ ms}^{-1}$

$\tan\theta = \frac{5}{15}$

$\theta = \tan^{-1}\left(\frac{5}{15}\right) = 18.4^\circ$

The resultant velocity is 15.8 ms^{-1} at S72°E.

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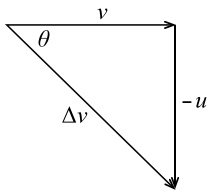
7. PHYSICS, M1 EQ-Bank 10801

$v = 50\ \text{kmh}^{-1}$ east, $u = 50\ \text{kmh}^{-1}$ north.

$\Delta v = v - u = v + (-u)$, where $-u = 50\ \text{kmh}^{-1}$ south



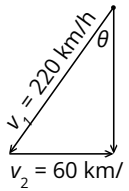
$$\Delta v = \sqrt{50^2 + 50^2} = \sqrt{5000} = 70.7\ \text{kmh}^{-1}$$

$$\tan\theta = \frac{50}{50}$$

$$\theta = \tan^{-1}\left(\frac{50}{50}\right) = 45^\circ$$

Change in velocity of the car = 70.7 kmh^{-1}, S45°E.

8. PHYSICS, M1 EQ-Bank 11602



$$\sin\theta = \frac{v_2}{v_1}$$

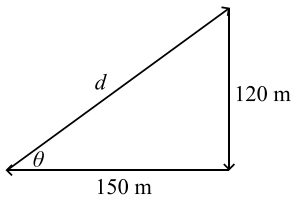
$$\theta = \sin^{-1}\left(\frac{60}{220}\right) = 16^\circ$$

$\therefore$ The pilot must steer the helicopter $\text{S}16^\circ\text{W}$.

9. PHYSICS, M1 EQ-Bank 10623

a. $d = v \times t = 5 \times 30 = 150\ \text{m}$

b. Stage 1 and Stage 2 displacement diagram:



Using Pythagoras:

$$d = \sqrt{150^2 + 120^2} = \sqrt{36900} = 192.1\ \text{m}$$

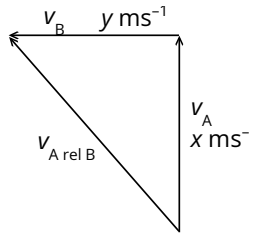
$$\tan\theta = \frac{120}{150}$$

$$\theta = \tan^{-1}\left(\frac{120}{150}\right) = 38.7^\circ$$

$\therefore$ Displacement of the start point from the runner is 192.1 m, N51.3°E.

10. PHYSICS, M1 EQ-Bank 11513

$v_{\text{A rel B}} = v_A - v_B = v_A + (-v_B)$



$$v_{\text{A rel B}} = \sqrt{x^2 + y^2}$$

Double the velocities of Boat A and Boat B – new relative velocity is:

$$\sqrt{(2x)^2 + (2y)^2} = \sqrt{4(x^2 + y^2)} = 2\sqrt{x^2 + y^2} = 2 \times v_{\text{A rel B}}$$

$\therefore$ The magnitude of the velocity of Boat A relative to Boat B is doubled when the boats double their speed.

11. PHYSICS, M1 EQ-Bank 10267

$s = 50,\ a = 9.8,\ u = -15$

Downwards $\Rightarrow$ positive direction

$$s = ut + \frac{1}{2}at^2$$

$$50 = -15t + \frac{1}{2} \times 9.8 \times t^2$$

$$0 = 4.9t^2 - 15t - 50$$

$$t = \frac{15 \pm \sqrt{225 - 4 \times 9.8 \times -50}}{2 \times 4.9}$$

$$= 5.07, -2.01$$

Time for the pen to fall = 5.07 seconds $(t > 0)$.

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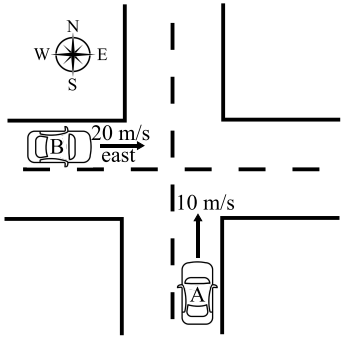
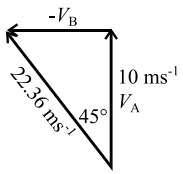
12. PHYSICS, M1 EQ-Bank 10445

- Velocity of A relative to B = $v_A - v_B = v_A + -v_B$



- The direction of $-v_B$ must be west, and therefore the direction of v_B is east.
- Using Pythagoras theorem, the magnitude of $-v_B$ is:

  $-v_B = \sqrt{22.36^2 - 10^2} = 20\ \text{ms}^{-1}$
- Thus the vector v_B is $20\ \text{ms}^{-1}$ east.



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